Exercise 2

On finding the term containing x^r in the expansion of binomial

From the school book | Understand | Apply | Higher Order Thinking Skills

1 Find :

(1) The coefficient of x^8 in the expansion of : $\left(x+\frac{2}{x^2}\right)^{14}$ « 364 »

(2) The coefficient of x in the expansion of : $\left(\frac{3x}{2}+\frac{2}{3x}\right)^{11}$ « 693 »

(3) The term containing x^{-8} in the expansion of : $\left(x-\frac{1}{2x^3}\right)^{12}$ « $-\frac{99}{4}x^{-8}$ »

(4) The term free of x in the expansion of : $\left(2x^2-\frac{1}{x}\right)^{12}$ « 7920 »

(5) The coefficient of $\left(\frac{x}{y}\right)^4$ in the expansion of : $\left(\frac{2x}{y}+\frac{y}{2x}\right)^{10}$ « 1920 »

(6) The coefficient of x^{12} in the expansion of : $x^2\left(\frac{x^2}{2}+\frac{2}{x^2}\right)^{15}$ « $\frac{3003}{32}$ »

(7) The coefficient of $\frac{1}{x^5}$ in the expansion of : $\frac{1}{x^3}\left(x+\frac{1}{x^2}\right)^{10}$ « 210 »

(8) The coefficient of x^9 in the expansion :

$$\left[(2+x)^{11}+11(2+x)^{10}(1+x)+\frac{11\times10}{2\times1}(2+x)^9(1+x)^2+\cdots+(1+x)^{11}\right]$$

« 253440 »

(9) The term containing x^6 in the expansion of : $\left(x^2+\frac{1}{x^2}\right)^{15}+\left(x+\frac{1}{x}\right)^{10}$ « $5050x^6$ »

(10) The term free of x in the expansion of $\left(x+\frac{1}{x}\right)^6-\left(x-\frac{1}{x}\right)^6$ « 40 »

2 Choose the correct answer from those given :

(1) The coefficient of the term containing x^3 in the expansion of : $(1+x)^{10}$ according to the ascending exponents of x = ...

(a) ${}^{10}C_2$ (b) ${}^{10}C_3$ (c) ${}^{10}C_4$ (d) 3

(2) *(2nd Session 2024)* If a x^{-3} is one of terms of the expansion of $\left(x^2+\frac{2}{x^3}\right)^{11}$, then a = ...

(a) 1848 (b) 3696 (c) 14784 (d) 29568

(3) *(2nd Session 2023)* The term free of x in the expansion of $\left(x-\frac{1}{x}\right)^8$ is .....

(a) T_3 (b) T_4 (c) T_5 (d) T_6

(4) *(Speciman 2025)* The term free of x in the expansion of : $\left(7x + \frac{6}{x^2}\right)^{12}$ according to the descending exponents of x is .....

(a) T_5 (b) T_6 (c) T_7 (d) T_8

(5) The coefficient of x in the expansion of : $\left(x^2 + \frac{a}{x}\right)^5$ is .....

(a) $9\,a^2$ (b) $10\,a^3$ (c) $10\,a^2$ (d) $10\,a$

(6) *(1st Session 2022)* If the fifth term in the expansion of $\left(2x - \frac{1}{x^2}\right)^n$ according to the descending exponents of x is the term free of x , then n = ...

(a) 4 (b) 5 (c) 12 (d) 15

(7) *(Trial 2023)* In the expansion of $\left(2x^2 + \frac{1}{2x}\right)^{3n}$ according to the descending exponents of x , if T_9 is the term free of x , then the coefficient of x^3 equals .....

(a) 495 (b) 198 (c) 792 (d) $\frac{495}{16}$

(8) *(1st Session 2023)* If the seventh term in the expansion of $\left(\frac{m}{x^2} - \frac{x^2}{m}\right)^{3n}$ according to the ascending exponents of x is the term free of x (where $m \in \mathbb{R}^+$) , then the value of this term .....

(a) – 12 (b) 924 (c) 4 m (d) $-924\,m^{12}$

(9) If the term free of x in the expansion of : $\left(\frac{a}{x^2} + x\right)^9$ is 672 , then a = ...

(a) 8 (b) 2 (c) 3 (d) 4

(10) If the absolute term in the expansion of : $\left(\sqrt{x} - \frac{k}{x^2}\right)^{10}$ equals 405 , then k = ...

(a) ± 1 (b) ± 2

(c) ± 3 (d) Nothing of the previous.

(11) *(Trial 2023)* In the expansion of : $x^3\,(1 + x)^7$ the coefficient of the term containing x^4 is .....

(a) ${}^7C_7$ (b) ${}^7C_4$ (c) ${}^7C_2$ (d) ${}^7C_1$

(12) In the expansion of : $(1 + x)^n$ according to ascending exponents of x , if the coefficient of x^5 equals the coefficient of T_{10} , then n = ...

(a) 9 (b) 5 (c) 14 (d) 4

(13) *(1st Session 2022)* In the expansion of $\left(\frac{1}{x} + k\,x^2\right)^8$ according to the ascending exponents of x , if the coefficient of $x^7 = 56$, then k = ...

(a) $\frac{-1}{5}$ (b) 5 (c) – 1 (d) 1

(14) In the expansion of : $\left(a\,x + \frac{1}{b\,x}\right)^{10}$ according to the descending exponents of x , if the term free of x equals the coefficient of the seventh term , then .....

(a) $ab = \frac{6}{5}$ (b) $ab = \frac{5}{6}$ (c) $ab = \frac{36}{25}$ (d) $ab = \frac{25}{36}$

(15) If the coefficients of x^3 and x^6 in the expansion of : $\left(a\,x^2 + \frac{1}{a\,x}\right)^9$ are equal , then a = ...

(a) 1 (b) – 1 (c) ± 1 (d) ± 2

(16) In the expansion of : $\left(2 + \frac{x}{3}\right)^n$ if the coefficients of x^7 and x^8 are equal , then n = ...

(a) 56 (b) 55 (c) 45 (d) 15

(17) In the expansion of : $\left(\sqrt[6]{x} - \frac{1}{\sqrt[3]{x}}\right)^9$ the term free of x equals .....

(a) 84 (b) – 84 (c) 8.4 (d) – 0.84

(18) *(1st Session 2021)* In the expansion of : $\left(x^5 - \frac{k}{x^2}\right)^{7n}$ according to the descending exponents of x , the term free of x is ..... , where k , $n \in \mathbb{Z}^+$

(a) T_{5n} (b) T_{5n+1} (c) T_{6n+1} (d) T_{6n-1}

(19) *(Speciman 2025)* If $(y^3 + x^4)^n = \cdots + k\,x^{12}\,y^{12} + \cdots$, then k = ...

(a) 56 (b) 48 (c) 42 (d) 35

(20) *(1st Session 2025)* If the expansion of $(x^3 + y^2)^n$ according to the descending exponents of x contains a term containg $x^{12}\,y^{20}$, then the middle term is .....

(a) T_7 (b) T_8 (c) T_{11} (d) T_{15}

(21) *(1st Session 2021)* In the expansion of : $\left(x^2 + 2 + \frac{1}{x^2}\right)^6$ the coefficient of the term containing x^2 is .....

(a) ${}^{12}C_5$ (b) ${}^{12}C_6$ (c) ${}^{12}C_2$ (d) ${}^{6}C_5$

(22) *(Trial 2021)* In the expansion of $\left(a\,x^2 - \frac{b}{x}\right)^{12}$ according to the descending exponents of x , then T_7 is .....

(a) the term containing x^6 (b) the term free of x

(c) the term before the last. (d) the term containing x^7

(23) *(1st Session 2024)* In the expansion of $(2 + x)^{10}$ the sum of the coefficients of the terms containing x^k such that k is divisible by 4 equals .....

(a) 13620 (b) 13622 (c) 14644 (d) 15904

(24) *(1st Session 2024)* In the expansion of $(a + x)^n$ if the term free of x and the coefficient of x are 8 , 24 respectively , then a : n = ...

(a) 3 : 1 (b) 1 : 3 (c) 1 : 2 (d) 2 : 1

(25) The coefficients of x^n , x^m in the expansion of : $(1 + x)^{m+n}$ .....

(a) are equal. (b) are equal but they have different sign.

(c) multiplicative inverse of each other. (d) one of them half the other.

(26) If $0 \leq k \leq n$, then the coefficient of x^k in the expansion :
$1 + (1 + x) + (1 + x)^2 + (1 + x)^3 + \cdots + (1 + x)^n$ is .....

(a) ${}^{n+1}C_{k+1}$ (b) ${}^{n}C_{k}$ (c) ${}^{n+1}C_{n+k}$ (d) ${}^{n}C_{n-k+1}$

(27) The value of the term free of x in the expansion of : $\left(x + \frac{1}{x}\right)^{2n} + \left(x - \frac{1}{x}\right)^{2n}$
equals ..... (where n is odd)

(a) zero (b) $-2\,{}^{2n}C_{r}$ (c) $2\,{}^{2n}C_{n}$ (d) ${}^{2n}C_{n}$

(28) *(Speciman 2025)* The coefficient of x^5 in the expansion of : $(1 + x^2)^5 (1 + x)^4$
equals .....

(a) 36 (b) 49 (c) 60 (d) 126

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3 In the expansion of : $\left(2x^2 + \frac{\ell}{x^3}\right)^{10}$, if the coefficient of x^5 equals the coefficient of x^{15} , find the value of ℓ where ℓ is positive. « $\frac{\sqrt{3}}{3}$ »

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4 In the expansion of : $\left(\frac{2x}{3} + \frac{3}{2x^2}\right)^{12}$, find each of :

(1) Middle term.

(2) Term containing x^3 « $924\,x^{-6}$, $\frac{14080}{729}x^3$ »

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5 In the expansion of : $\left(9x^2 + \frac{1}{3x}\right)^{9}$, find the value of the term free of x , then prove that the two middle terms are equal when $x = \frac{1}{3}$ « 84 »

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6 In the expansion of : $\left(\frac{x}{2} - \frac{2}{x^3}\right)^{10}$, prove that : this expansion does not contain term free of x , then find the coefficient of $\frac{1}{x^2}$ « $-\frac{15}{2}$ »

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7 *(1st Session 2010)* In the expansion of : $\left(x^2 + \frac{1}{8x}\right)^{13}$ according to the descending exponents of x :

(1) Prove that there is no term of first degree

(2) If the fourth term and the eleventh term are equal , find the value of : x « $\frac{1}{2}$ »

8 *(1st Session 2008)* In the expansion of : $\left(\frac{5}{x}+x\right)^8$ according to the ascending exponents of x :

(1) Prove that : the term free of x is the middle term then find its value.

(2) Find the value of x that makes the ratio between the third term and the seventh term equals 1 : 16

« 43750 , $\pm\sqrt{10}$ »

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9 In the expansion of : $\left(x^2+\frac{1}{x}\right)^{15}$, find the ratio between the term free of x and the sum of the coefficients of the two middle terms.

« $\frac{7}{30}$ »

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10 In the expansion of : $\left(x^2+\frac{1}{a\,x}\right)^8$, if the coefficient of the middle term equals the coefficient of x^{10} , find the value of : a

« $\frac{\pm\sqrt{10}}{2}$ »

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11 In the expansion of : $\left(4\,x^2+\frac{1}{2\,x}\right)^{13}$, find the coefficient of x^5 then find the value of x which makes the two middle terms in this expansion are equal , and prove that this expansion has no term free of x

« 54912 , $\frac{1}{2}$ »

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12 Find the coefficient of x^6 in the expansion of : $\left(x+\frac{1}{x}\right)^9\left(x-\frac{1}{x}\right)^9$, prove that this expansion does not contain a term containing x^{-3}

« – 84 »

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13 If the expansion of : $(x^5+x^{-2})^{2\,n}$ contains a term free of x , prove that : n = 7 or multiple of 7 , then calculate this term when n = 7

« 1001 »

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14 If the order of the term free of x in the expansion of : $\left(2\,x^2-\frac{3}{x}\right)^{21}$ according to the descending exponents of x equals the order of the term free of x in the expansion of : $\left(x+\frac{3}{x}\right)^{2n}$ according to the descending exponents of x , find the value of : n

« 14 »

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15 If the seventh term in the expansion of : $\left(\frac{x^2}{\sqrt{2}}-\frac{\sqrt{2}}{x^2}\right)^n$ according to the descending exponents of x is the term free of x , find the value of n then calculate the value of this term.

« n = 12 , 924 »

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16 In the expansion of : $\left(x^k+\frac{1}{x}\right)^{10}$ where : $K\in\mathbb{Z}^+$, **find :**

(1) The values of k that makes the expansion has a term free of x

(2) The ratio between the term free of x and the coefficient of the middle term , this is for the greatest value of k you got from first.

« $\frac{5}{126}$ »

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17 In the expansion of : $\left(x+\frac{1}{x}\right)^{2n}$, prove that the term free of x is the middle term , then find the value of this term when : n = 8

« 12870 »

18 Prove that the term free of x in the expansion of : $\left(x^2 + \frac{1}{x^3}\right)^{5n}$ where : $n \in \mathbb{Z}^+$ equals $\frac{\lfloor 5n}{\lfloor 2n \, \lfloor 3n}$

19 If the third term in the expansion of : $\left(2x + \frac{1}{x^2}\right)^n$ according to the descending exponents of x is free of x , find the value of y which makes this term equal to the second term in the expansion of $(1 + y^3)^{30}$ according to the ascending exponents of y « n = 6 , y = 2 »

20 If the coefficient of x^b , x^c in the expansion of : $(1 + x)^a$ according to ascending exponents of x are equal , prove that : a = b + c

21 Find the value of the term free of x in the expansion of : $(x^2 + 1)\left(x - \frac{1}{x}\right)^{10}$ « – 42 »

22 Find the coefficient of x^2 in the product of : $(1 - x)^8 (3 + 2x)^6$ « 1944 »

23 If the coefficient of x^2 in the expansion of : $(1 + 2x)^5 (1 + nx)^4$ is – 26 , find the value of : n « – 3 , $\frac{-11}{3}$ »

Higher skills

24 **Choose the correct answer from those given :**

(1) The coefficient of x^{20} in the expansion of : $(1 + x^2)^{40}\left(x^2 + 2 + \frac{1}{x^2}\right)^{-5}$ equals .....

(a) 1 (b) $^{30}C_{10}$ (c) $^{30}C_5$ (d) 30

(2) The coefficient of x^{13} in the expansion of : $\sum_{r=0}^{50} {}^{50}C_r (x - 1)^{50-r} \times 2^r$ equals .....

(a) $^{50}C_{13}$ (b) $^{50}C_{12}$ (c) 1 (d) zero

(3) The term free of x in the expansion of : $\left(1 - \frac{1}{x}\right)^8 (x + 1)^8$ is .....

(a) $-{}^8C_6$ (b) $^8C_5$ (c) $-{}^8C_4$ (d) $^8C_4$

(4) If A , B are the coefficients of x^n in the expansion of : $(1 + x)^{2n}$, $(1 + x)^{2n-1}$ respectively , then .....

(a) A = 3 B (b) B = 2 A (c) A = 2 B (d) B = 3 A

(5) The coefficients of x^6 in the expansion of : $\left[(1 + x)^6 + (1 + x)^7 + (1 + x)^8 + \cdots + (1 + x)^{15}\right]$ equals .....

(a) $^{16}C_9$ (b) $^{16}C_5 - {}^5C_4$ (c) $^{16}C_7 - 1$ (d) $^{15}C_6$